

Due April 7 at the start of class 1:50 pm

Submit the following.

- Complete pages HW – 56 thru HW - 58.
- Work on the MML assignments.

7. The sides of a square *increase* in length at a constant rate of 2 in/s. At what rate is the length of the diagonal of the square changing when the sides are 5 in long?

Give an interpretation of the answer.

- Step 1: Draw a sketch and label the variable of interest. Next, write an equation that represents the quantity that is required to find the solution to the problem

- step 2: Rewrite the equation in step 1 to show the explicit functions.

- step 3: Use the Chain rule on the function equation.

- step 4: Define all numerical values from the problem statement using the correct variable and derivative notation. Give the derivative that the problem requires us to find in correct derivative notation.

- step 5: Substitute all numerical values into the derivative equation.

- step 6: Solve for the unknown derivative.

8. The volume of a cylinder is given by $V = \pi r^2 h$. The height, h , of the cylinder is a constant 10 cm. When the radius, r , of the cylinder is 2 cm, the radius is decreasing at a rate of 5 cm/s. At what rate is the volume changing? Give an interpretation of the answer.

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- step 1: Rewrite the equation to show the explicit functions.

- step 2: Use the Chain rule on the function equation from step 1:

- step 3: Define all numerical values from the problem statement using the correct variable and derivative notation. Include units for all quantities. State the derivative that the problem requires us to find in correct derivative notation.

- step 4: Substitute all numerical values into the derivative equation.

- step 5: Solve for the unknown derivative.

9. The surface area of a cylinder is given by $S = 2\pi rh + 2\pi r^2$. The height, h , of the cylinder is a constant 8 cm and the radius, r , of the cylinder is decreasing at a rate of 3 cm/s. At what rate is the surface area changing when the radius of the cylinder is 2 cm? Give an interpretation of the answer.

10. The volume of a cube is given by $V = x^3$. If the volume of the side of the cube is decreasing at 5 cm³/s, at what rate is the volume changing when $x = 10$ cm?